

Architecture — yanatech.co.uk Homelab

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Cluster version: Kubernetes v1.32.13 (kubeadm) — Ubuntu
24.04.4 LTS — containerd 2.2.4

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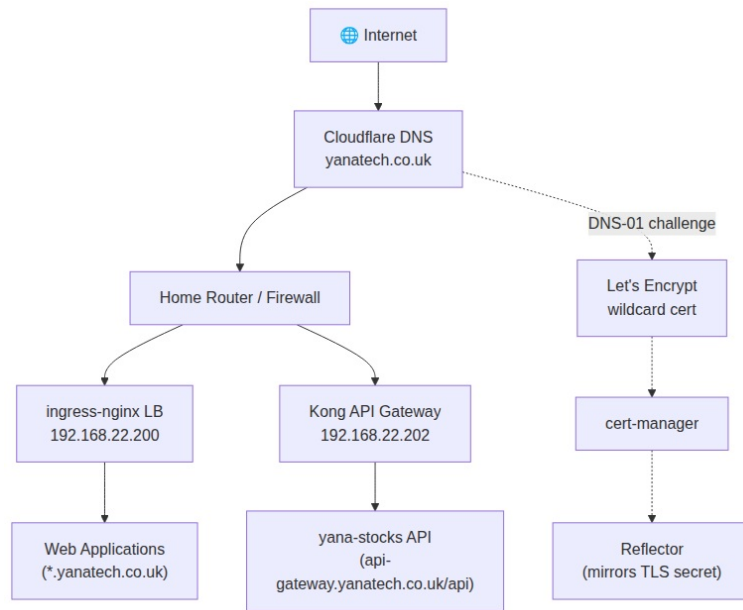
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1. Overview

A self-hosted, production-grade homelab running on a 6-node Kubernetes cluster across 3 Proxmox hypervisors. The cluster is managed entirely via GitOps (ArgoCD) — nothing is deployed manually without a subsequent commit to github.com/akann/k8s-apps.

Design principles:

- GitOps-first: git is the single source of truth
- Zero-trust networking: every namespace has default-deny-all NetworkPolicy
- Secrets never in git: all secrets pulled from InFisical via External Secrets Operator
- Single wildcard TLS cert reflected to all namespaces via Reflector
- SSO on all web UIs via Authentik



diagram

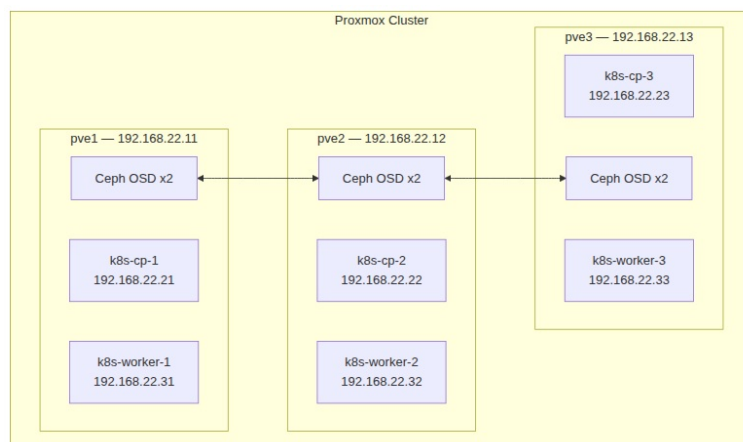
2. Physical Infrastructure

2.1 Proxmox Hypervisors

Host	IP	Role
pve1	192.168.22.11	Proxmox hypervisor + Ceph OSD
pve2	192.168.22.12	Proxmox hypervisor + Ceph OSD
pve3	192.168.22.13	Proxmox hypervisor + Ceph OSD

All Kubernetes VMs run as Proxmox guests. Ceph monitors are co-located on the Proxmox hosts at 192.168.22.11-13:6789.

2.2 Physical Topology



diagram

3. Kubernetes Cluster

3.1 Nodes

Node	Role	IP	OS	K8s Version	CRI
k8s-cp-1	control-plane	192.168.22.21	Ubuntu 24.04.4 LTS	v1.32.13	containerd 2.2.4
k8s-cp-2	control-plane	192.168.22.22	Ubuntu 24.04.4 LTS	v1.32.13	containerd 2.2.4
k8s-cp-3	control-plane	192.168.22.23	Ubuntu 24.04.4 LTS	v1.32.13	containerd 2.2.4
k8s-worker-1	worker	192.168.22.31	Ubuntu 24.04.4 LTS	v1.32.13	containerd 2.2.4
k8s-worker-2	worker	192.168.22.32	Ubuntu 24.04.4 LTS	v1.32.13	containerd 2.2.4
k8s-worker-3	worker	192.168.22.33	Ubuntu 24.04.4 LTS	v1.32.13	containerd 2.2.4

The cluster uses **stacked etcd** — etcd runs on each control-plane node alongside the API server. kubeadm manages the cluster lifecycle.

3.2 Namespaces

Namespace	Purpose
actions-runner	Self-hosted GitHub Actions runners (ARC)
apicurio	API schema registry
argo-rollouts	Canary/blue-green deployment controller
argocd	GitOps engine
authentik	SSO / identity provider
ceph-csi-rbd	Ceph RBD CSI driver
cert-manager	TLS certificate automation
cilium-secrets	Cilium mTLS secrets
cnpg-clusters	CloudNativePG shared PostgreSQL cluster
cnpg-system	CloudNativePG operator
external-secrets	External Secrets Operator
goldilocks	VPA resource recommendations
gotify	Push notification server
harbor	Container image registry
headlamp	Kubernetes web UI
immich	Photo management (self-hosted Google Photos)
infisical	Secrets manager
ingress-nginx	HTTP/S ingress controller
kafka	Strimzi Kafka cluster + operator

keda	Kubernetes Event-Driven Autoscaling
kong	API gateway
kube-system	Core system components, Cilium, Hubble
kured	Node reboot daemon (kured)
metallb-system	Bare-metal load balancer
minio	S3-compatible object storage
mongo-express	MongoDB web UI
mongodb	MongoDB replicaset
monitoring	Prometheus, Grafana, Alertmanager, Loki, Tempo
nextcloud	Self-hosted cloud storage
pgadmin	PostgreSQL web UI
redis	Redis standalone
redis-insight	Redis web UI
reflector	Kubernetes secret/configmap reflector
reloader	Automatic pod rolling restarts on config changes
uptime-kuma	Uptime monitoring
vaultwarden	Bitwarden-compatible password manager
velero	Cluster backup
yana-stocks	yana-stocks microservices application
yanatech	Static site / landing page

4. Networking

4.1 CNI — Cilium

Mode: Native routing (no encapsulation — pods route directly at L3 across nodes).

Key implications:

- Inter-pod traffic is not encapsulated → lower overhead, full observability via Hubble
- NetworkPolicy objects work, but **Ceph OSD egress** and **Grafana → Prometheus** require CiliumNetworkPolicy with toCIDR — standard NetworkPolicy ClusterIP routing fails in native routing mode
- ESO webhook disabled because kube-apiserver cannot connect to in-cluster services via node IPs without a CiliumNetworkPolicy

Hubble UI: hubble.yanatech.co.uk (Authentik-protected)

4.2 Load Balancer — MetalLB

MetalLB provides LoadBalancer type services in bare-metal mode.

VIP	Service
192.168.22.200	ingress-nginx (all web traffic)
192.168.22.201	infisical bundled nginx (scaled to 0, do not use)
192.168.22.202	Kong API Gateway (kong-gateway-proxy)

Pool: 192.168.22.200–249

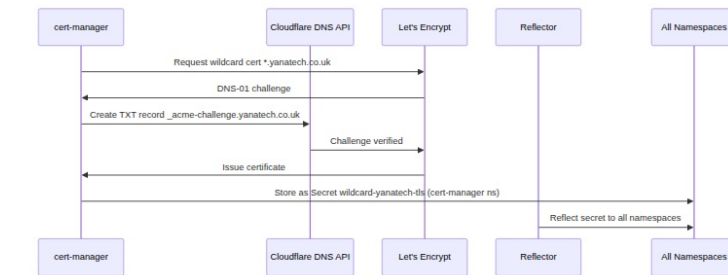
4.3 Ingress Controllers

Two ingress classes are active:

Class	Controller	VIP	Purpose
nginx	ingress-nginx	192.168.22.200	All web UIs, TLS termination
kong	Kong Ingress Controller	192.168.22.202	yana-stocks API routing, JWT validation
infisical-nginx	infisical bundled nginx	192.168.22.201	Infisical internal (disabled)

Note: ingress-nginx requires `allowSnippetAnnotations: true` and `annotations-risk-level: Critical` for Authentik forward-auth auth-snippet annotations.

4.4 TLS



diagram

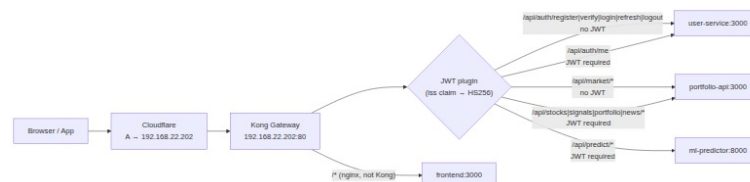
- **ClusterIssuer:** letsencrypt-prod (Ready)
- **Secret name:** wildcard-yanatech-tls (reflected to every namespace by Reflector)
- **Challenge:** Cloudflare DNS-01 (no port 80 exposure required)

4.5 Traffic Flow — Web Request



diagram

4.6 Traffic Flow — API Request (yana-stocks)



diagram

4.7 SSO — Authentik Forward Auth

For apps without native OIDC, Authentik forward auth is used:

1. App's Ingress has `nginx.ingress.kubernetes.io/auth-url` pointing to the app's outpost endpoint
2. Each protected app gets an `ak-outpost-<name>` pod deployed automatically in the `authentik` namespace
3. An `ExternalName` Service in the app's namespace routes `/outpost.goauthentik.io` to the outpost
4. A second Ingress in the app namespace routes the outpost path

Important: `auth-url` must use the **external** hostname (not internal service IP) — outposts match by external host.

4.8 Network Policies

Every namespace has `default-deny-all`. Key policy files:

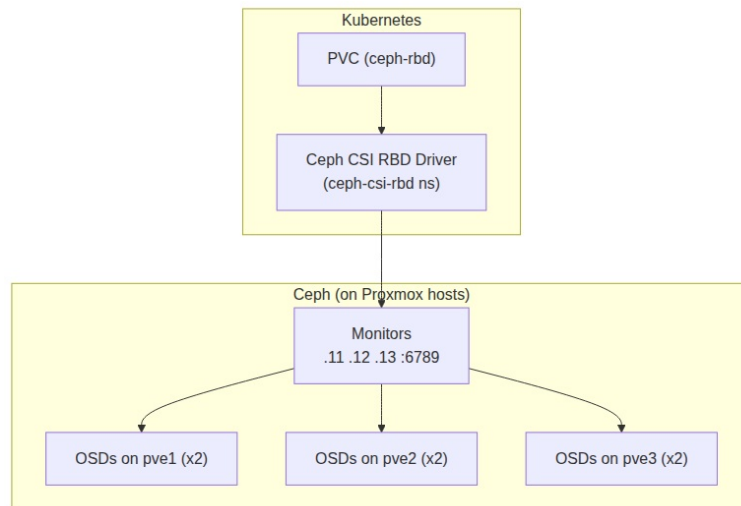
File	Coverage
<code>infrastructure/network-policies/netpol-infrastructure.yaml</code>	All infrastructure namespaces
<code>infrastructure/network-policies/netpol-cnpg.yaml</code>	CNPG operator + clusters
<code>infrastructure/network-policies/netpol-monitoring.yaml</code>	Monitoring stack
<code>infrastructure/network-policies/netpol-apiserver-egress.yaml</code>	kube-apiserver egress for operator namespaces
<code>infrastructure/cilium/ciliumnetpol-ceph-osd.yaml</code>	Ceph OSD egress (ports 6802-6809, toCIDR 192.168.22.0/24)
<code>infrastructure/cilium/ciliumnetpol-grafana-prometheus.yaml</code>	Grafana → Prometheus (ClusterIP bypass)

5. Storage

5.1 Ceph RBD

Property	Value
Type	Ceph RBD (block storage)
StorageClass	<code>ceph-rbd</code> (default)
Raw capacity	8.4 TiB
OSDs	6 (2 per Proxmox host)
Monitors	192.168.22.11-13:6789
Cluster ID	92197a62-7cf9-49eb-a0cb-5e0b9bbff52a
Access mode	RWO (ReadWriteOnce)
CSI namespace	<code>ceph-csi-rbd</code>

All `PersistentVolumeClaims` cluster-wide use `ceph-rbd`. There is no NFS or local-path storage.



diagram

Critical: Ceph CSI egress to OSD ports 6802-6809 **must** use CiliumNetworkPolicy with toCIDR: 192.168.22.0/24 — standard NetworkPolicy cannot reach bare-metal IPs in Cilium native routing mode.

5.2 PVC Inventory

Namespace	PVC	Size	Consumer
cnpg-clusters	pg-main-1/2/4	50 Gi × 3	Shared PostgreSQL (pg-main)
harbor	harbor-registry	100 Gi	Image layers
harbor	harbor-database	10 Gi	Harbor PostgreSQL
harbor	harbor-redis / trivy	5 Gi / 10 Gi	Harbor cache/scanner
immich	immich-library	200 Gi	Photo library
immich	immich-postgres-1	20 Gi	Immich CNPG PostgreSQL
kafka	data-kafka-cluster-dual-role-{0,1,2}	20 Gi × 3	Kafka log data
minio	minio	50 Gi	Object storage
mongodb	datadir-mongodb-{0,1}	10 Gi × 2	MongoDB replicas
monitoring	prometheus-db	20 Gi × 2	Prometheus TSDB
monitoring	alertmanager-db	5 Gi × 2	Alertmanager state
monitoring	storage-loki-0	20 Gi	Loki log storage
monitoring	storage-tempo-0	20 Gi	Tempo trace storage
nextcloud	nextcloud-nextcloud	100 Gi	Nextcloud data
redis	redis-data-redis-master-0	5 Gi	Redis AOF/RDB

vana-stocks	user-service-pg-1	10 Gi	yana-stocks PostgreSQL
vaultwarden	vaultwarden-data	5 Gi	Vault data
uptime-kuma	uptime-kuma-pvc	5 Gi	Kuma database

6. Secret Management

6.1 Architecture



diagram

ESO webhook is disabled (webhook.create: false, certController.create: false) — Cilium native routing blocks kube-apiserver node IP connections to in-cluster webhook services.

6.2 ExternalSecret Pattern

```

apiVersion: external-secrets.io/v1
kind: ExternalSecret
metadata:
  name: my-secret
  namespace: my-namespace
spec:
  refreshInterval: 1h
  secretStoreRef:
    name: infisical
    kind: ClusterSecretStore
  target:
    name: my-secret
    creationPolicy: Owner
  data:
    - secretKey: my-key
      remoteRef:
        key: /my-folder/MY_SECRET_NAME
  
```

6.3 Secret Path Convention

Secrets in Infisical follow the path pattern /namespace/SECRET_NAME.

6.4 Bootstrap Process

For a fresh cluster, Vaultwarden (at vault.yanatech.co.uk) is the source of truth for manually provisioning the initial Infisical credentials and any secrets that ESO cannot yet pull.

7. GitOps & CI/CD

7.1 ArgoCD

Property	Value
Version	v3.4.2
URL	https://argocd.yanatech.co.uk
Source repo	https://github.com/akann/k8s-apps
Sync policy	Automated (prune: true, selfHeal: true)
Namespace creation	Via CreateNamespace=true syncOption
Apply method	ServerSideApply=true

All applications use **app-of-apps pattern** or direct ArgoCD Application manifests. The sync wave annotation controls deployment ordering.

7.2 ArgoCD Sync Waves

Wave	Components
0	MetalLB, Ceph CSI
1	Cilium, cert-manager, ingress-nginx
2	MetalLB IPAddressPool+L2Advertisement, cert-manager ClusterIssuers
3	Reflector, Reloader, Kured, Descheduler, KEDA, Argo Rollouts, NetworkPolicies
4	Authentik, kube-prometheus-stack, Tempo, Velero
5	Loki+Promtail, Headlamp, Goldilocks, Redis, MongoDB, MinIO, Kong
6	ESO, Infisical, Redis Insight, Mongo Express
7	CNPG operator, CNPG clusters (pg-main, immich-postgres)
8	Harbor, Actions Runner Controller
9+	Applications (Vaultwarden, Kafka, Immich, Nextcloud, etc.)
10+	yana-stocks services

7.3 yana-stocks CI/CD Pipeline



diagram

- **Turborepo** only builds services whose source changed (--filter=[HEAD^1])
- **Harbor** stores images tagged with git SHA + latest
- **Runners** are self-hosted ARC runners in the actions-runner namespace
- **Secret:** HARBOR_USERNAME, HARBOR_PASSWORD, GH_PAT in GitHub Actions

7.4 Known Permanent OutOfSync (cosmetic, all Healthy)

These apps show OutOfSync in ArgoCD UI but are functioning correctly:

App	Reason
actions-runner-	

controller	OCI registry limitation
argo-rollouts	Cluster-scoped CRDs tracked twice
infisical	Bundled nginx chart mutation
kafka	Strimzi bootstrap Service patch timeout (upstream)
immich	SharedResourceWarning (ingress shared between two apps)

8. Platform Services

8.1 Databases

CloudNativePG (CNPG) — PostgreSQL

Cluster	Namespace	Instances	Primary	Size	Consum
pg-main	cnpg-clusters	3	pg-main-1	50 Gi x 3	vaultwarden, authentik, nextcloud, infisical, apicur
immich-postgres	immich	1	immich-postgres-1	20 Gi	Immich (pgvector/Vecto
user-service-pg	yana-stocks	1	user-service-pg-1	10 Gi	yana-stocks us service

pg-main connection: pg-main-rw.cnpg-clusters.svc.cluster.local:5432

Immich: uses ghcr.io/tensorchord/cloudnative-vectorchord:16-1.1.1 (vector extension for AI search)

MongoDB

- **StatefulSet:** mongodb (2 replicas) + mongodb-arbiter (1)
- **ReplicaSet:** rs0
- **Connection:** mongodb-headless.mongodb.svc.cluster.local:27017
- **PVC:** 10 Gi per replica (ceph-rbd)
- **UI:** Mongo Express at mongo.yanatech.co.uk (Authentik-protected)

Redis

- **StatefulSet:** redis-master (standalone, 1 replica)
- **Connection:** redis-master.redis.svc.cluster.local:6379
- **PVC:** 5 Gi (ceph-rbd)
- **UI:** Redis Insight at redis.yanatech.co.uk (Authentik-protected)

MinIO

- **Connection:** minio.minio.svc.cluster.local:9000
- **PVC:** 50 Gi (ceph-rbd)
- **Console:** minio-console.yanatech.co.uk
- **API endpoint:** minio.yanatech.co.uk

8.2 Message Broker — Kafka (Strimzi)

Property	Value
Operator	Strimzi strimzi-cluster-operator
Cluster name	kafka-cluster
Mode	KRaft (combined broker + controller, no ZooKeeper)
Brokers	3 dual-role pods (kafka-cluster-dual-role-{0,1,2})
Bootstrap	kafka-cluster-kafka-bootstrap.kafka.svc.cluster.local:9092
PVC per broker	20 Gi (ceph-rbd)
UI	kafka-ui.yanatech.co.uk (Authentik-protected)

Topics:

Topic	Partitions	Replication	Retention	P
stocks.prices.raw	3	3	24h	pi in
stocks.prices.processed	3	3	7d	pi pi
stocks.signals.sentiment	3	3	7d	se ar
stocks.signals.prediction	3	3	7d	m pi
stocks.portfolio.events	3	3	30d	pe se

8.3 Container Registry — Harbor

- **URL:** harbor.yanatech.co.uk
- **Project:** yana-stocks
- **Image format:** harbor.yanatech.co.uk/yana-stocks/<service>:<tag>
- **Storage:** 100 Gi registry PVC + 10 Gi database PVC (ceph-rbd)
- **Vulnerability scanning:** Trivy (10 Gi PVC)

8.4 Cluster Maintenance

Tool	Purpose	Namespace
Kured	Auto-reboot nodes when /var/run/reboot-required is set (after kernel updates)	kured
Descheduler	Evict and reschedule pods to rebalance cluster	kube-system (daemonset)
Goldilocks	VPA-based resource request recommendations	goldilocks
Reloader	Rolling restart pods when ConfigMaps/Secrets change	reloader
Reflector	Mirror wildcard-yanatech-tls Secret + any annotated resources across namespaces	reflector

9. Applications

9.1 Service URLs

Service	URL	Auth	No
ArgoCD	https://argocd.yanatech.co.uk	Native	GitOps U
Authentik	https://authentik.yanatech.co.uk	Native	SSO IdP
Grafana	https://grafana.yanatech.co.uk	Authentik OIDC	Metrics dashboar
Immich	https://photos.yanatech.co.uk	Native	Photo managen
Infisical	https://infisical.yanatech.co.uk	Native	Secrets r
Harbor	https://harbor.yanatech.co.uk	Native	Containe registry
Nextcloud	https://cloud.yanatech.co.uk	Native	Cloud stc
Kong API GW	https://api- gateway.yanatech.co.uk	JWT (per- route)	yana-stoc
MinIO Console	https://minio- console.yanatech.co.uk	Native	Object st
MongoDB UI	https://mongo.yanatech.co.uk	Authentik forward	Mongo E
Redis UI	https://redis.yanatech.co.uk	Authentik forward	Redis Ins
Headlamp	https://headlamp.yanatech.co.uk	SA token	Kubernel
Kafka UI	https://kafka-ui.yanatech.co.uk	Authentik forward	Kafka br
Argo Rollouts	https://rollouts.yanatech.co.uk	Authentik forward	Canary dashboar
pgAdmin	https://pgadmin.yanatech.co.uk	Native	PostgreS
Apicurio	https://apicurio.yanatech.co.uk	Authentik forward	Schema 1
Vaultwarden	https://vault.yanatech.co.uk	Native	Password manager
Uptime Kuma	https://status.yanatech.co.uk	Authentik forward	Uptime monitori
Hubble UI	https://hubble.yanatech.co.uk	Authentik forward	Cilium ne viz
Goldilocks	https://goldilocks.yanatech.co.uk	Authentik forward	Resource recomme
Gotify	https://gotify.yanatech.co.uk	Native	Push not
yana-stocks	https://stocks.yanatech.co.uk	JWT (in- app)	Stock ma
yanatech	https://yanatech.co.uk	Public	Landing

Note on Headlamp: Authentik SSO is broken upstream; use a long-lived ServiceAccount token:

```
kubectl create token headlamp -n headlamp --duration=8760h
```

10. yana-stocks Architecture

A production-grade, event-driven microservices application for real-time stock data, portfolio management, sentiment analysis, and ML price prediction.

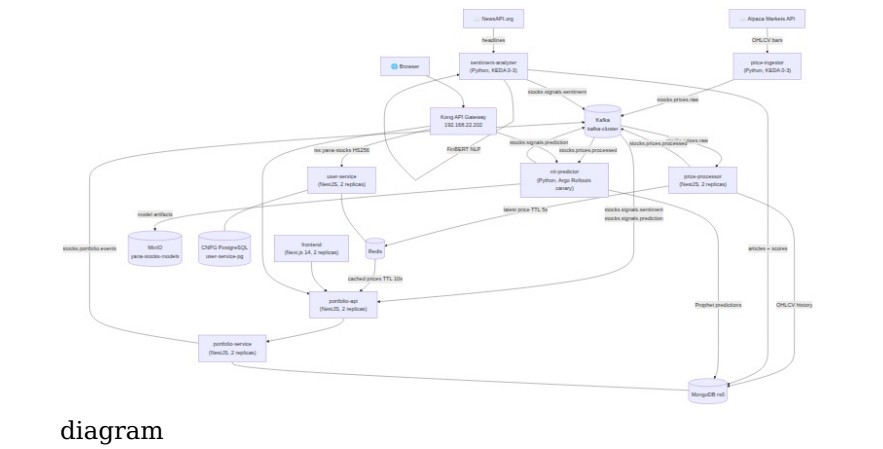
Source: github.com/akann/yana-stocks (Turborepo monorepo)

Namespace: yana-stocks

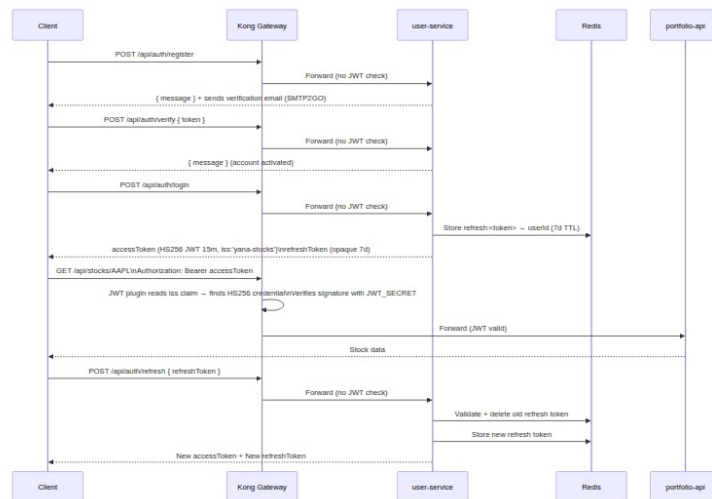
10.1 Service Overview

Service	Language	Pattern	Replicas	Dependencies
frontend	Next.js 14 (App Router)	Deployment	2	portfolio-api
portfolio-api	NestJS	Deployment	2	user-service, portfolio-service, price-processor, ml-predictor, Redis
portfolio-service	NestJS	Deployment	2	MongoDB, Kafka
price-processor	NestJS	Deployment	2	MongoDB, Redis, Kafka
user-service	NestJS	Deployment	2	CNPG PostgreSQL, Redis, Kafka
price-ingestor	Python	Deployment + KEDA	0-3	Alpaca API, Kafka
sentiment-analyzer	Python	Deployment + KEDA	0-3	NewsAPI, MongoDB, Kafka
ml-predictor	Python	Argo Rollouts (canary)	1	MongoDB, MinIO, Kafka

10.2 Data Flow



10.3 Authentication Flow



diagram

10.4 Kong API Routes

All routes have cors plugin. JWT-required routes additionally have jwt-auth plugin (validates HS256 signature using iss: 'yana-stocks' credential).

Route	PathType	Target	JWT Required
/api/auth/register, /api/auth/verify, /api/auth/login	Exact	user-service:3000	No
/api/auth/refresh, /api/auth/logout	Exact	user-service:3000	No
/api/auth/me	Exact	user-service:3000	Yes
/api/market/*	Prefix	portfolio-api:3000	No
/api/stocks/*	Prefix	portfolio-api:3000	Yes
/api/signals/*	Prefix	portfolio-api:3000	Yes
/api/portfolio/*	Prefix	portfolio-api:3000	Yes
/api/news/*	Prefix	portfolio-api:3000	Yes
/api/predict/*	Prefix	ml-predictor:8000	Yes
/*	Prefix	frontend:3000	No (nginx)

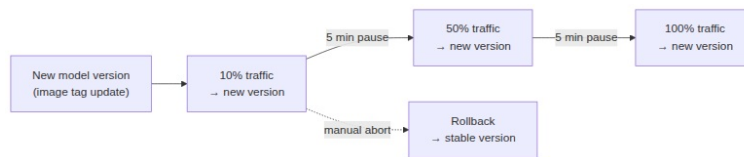
Note: /api/portfolio/* is handled by portfolio-api, which internally proxies to portfolio-service. Kong never routes directly to portfolio-service. The /* frontend route uses ingress-nginx (not Kong).

10.5 KEDA Autoscaling

Both price-ingestor and sentiment-analyzer scale to zero when their Kafka consumer group has no lag:

```
# ScaledObject pattern (both services)
triggers:
  - type: kafka
  metadata:
    bootstrapServers: kafka-cluster-kafka-
bootstrap.kafka.svc.cluster.local:9092
    consumerGroup: <service-name>
    topic: <input-topic>
    lagThreshold: "10"
minReplicaCount: 0
maxReplicaCount: 3
```

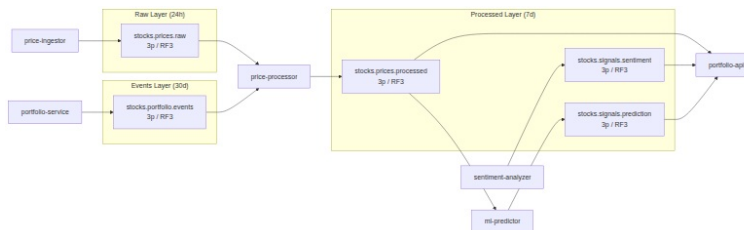
10.6 Argo Rollouts — ml-predictor Canary



diagram

Dashboard: rollouts.yanatech.co.uk (Authentik-protected)

10.7 Kafka Topic Flow



diagram

10.8 Monorepo Structure

yana-stocks/	# github.com/akann/yana-stocks
├─ apps/	
│ ├─ frontend/	# Next.js 14 (App Router),
│ │ TailwindCSS, Recharts	
│ │ ├─ price-processor/	# NestJS, Mongoose, ioredis, KafkaJS
│ │ ├─ user-service/	# NestJS, Prisma, JWT, KafkaJS
│ │ ├─ portfolio-service/	# NestJS, Mongoose, KafkaJS
│ │ ├─ portfolio-api/	# NestJS, aggregator + auth proxy
│ │ └─ e2e/	# Playwright (Chromium + iPhone 14)
├─ services/	
│ └─ price-ingestor/	# Python 3.12, confluent-kafka,
├─ alpaca-py, uv	
│ └─ sentiment-analyzer/	# Python 3.12, HuggingFace FinBERT,
├─ NewsAPI, uv	
│ └─ ml-predictor/	# Python 3.12, FastAPI, Prophet,
├─ scikit-learn, uv	
└─ packages/	
└─ shared-types/	# TypeScript interfaces (Stock,

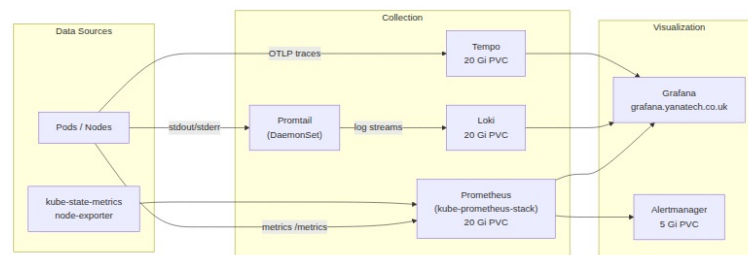
```

OHLCV, Portfolio, ...)
├── shared-dto/           # Validation DTOs (class-validator)
├── kafka-client/         # KAFKA_TOPICS constants + config
factory
├── typescript-config/    # Shared tsconfig bases (base,
nextjs, nextjs)
├── eslint-config/        # Shared ESLint
└── prettier-config/      # Shared Prettier

```

11. Observability

11.1 Stack



diagram

Note: Grafana → Prometheus requires CiliumNetworkPolicy — ClusterIP routing for Prometheus fails with Cilium native routing, causing context deadline exceeded errors. See [infrastructure/cilium/ciliumnetpol-grafana-prometheus.yaml](#).

11.2 Network Observability

- **Hubble UI:** hubble.yanatech.co.uk — real-time L3/L4 flow visualization built into Cilium
 - **Authentik-protected** via forward auth
-

12. Deployment Patterns

12.1 Standard Deployment (most services)

```

apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: my-app
  namespace: argocd
  annotations:
    argocd.argoproj.io/sync-wave: "9"
spec:
  project: default
  source:
    repoURL: https://github.com/akann/k8s-apps
    path: apps/my-app
    targetRevision: HEAD
  destination:
    server: https://kubernetes.default.svc

```

```

    namespace: my-namespace
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
  syncOptions:
    - CreateNamespace=true
    - ServerSideApply=true

```

Use valuesObject: (not values: |) for Helm values to avoid YAML indentation issues.

12.2 Helm Chart Deployment

For charts hosted on OCI or HTTP registries:

```

spec:
  source:
    chart: <chart-name>
    repoURL: https://charts.example.com
    targetRevision: 1.2.3
  helm:
    valuesObject:
      key: value

```

12.3 Authentik Forward Auth Pattern

For apps without native OIDC (Kafka UI, Redis Insight, etc.):

```

# Main app Ingress
annotations:
  nginx.ingress.kubernetes.io/auth-url:
"https://<hostname>/outpost.goauthentik.io/auth/nginx"
  nginx.ingress.kubernetes.io/auth-signin:
"https://<hostname>/outpost.goauthentik.io/start?
rd=$escaped_request_uri"
  nginx.ingress.kubernetes.io/auth-response-headers: "Set-Cookie,X-
authentik-username,X-authentik-groups,X-authentik-email,X-authentik-
name,X-authentik-uid"
  nginx.ingress.kubernetes.io/auth-snippet: |
    proxy_set_header X-Original-URL
    $scheme://$http_host$request_uri;

```

Requires allowSnippetAnnotations: true and annotations-risk-level: Critical on ingress-nginx.

12.4 Adding a New Namespace Checklist

1. Add default-deny-all NetworkPolicy to appropriate netpol file
2. Add specific ingress/egress policies
3. Add allow-kube-apiserver-egress to netpol-apiserver-egress.yaml if namespace has operators/controllers
4. If the app needs TLS: the wildcard-yanatech-tls secret will be auto-reflected by Reflector (no action needed if annotation is on the namespace)
5. If secrets needed: create ExternalSecret resources pointing to the infisical ClusterSecretStore
6. If Authentik SSO needed: create Authentik Provider + Application + outpost; add auth annotations to Ingress

13. Backup & Recovery

13.1 Velero

Property	Value
Schedule	Weekly, Sundays at 02:00 UTC
Schedule name	velero-weekly-backup
Namespace	velero

Velero backs up Kubernetes resources and PVC snapshots. Configure a BackupStorageLocation pointing to MinIO (minio.minio.svc.cluster.local:9000) for object storage of backups.

13.2 CNPG Backup

CloudNativePG clusters support WAL archiving to S3-compatible storage. Configure backup targets per-cluster to stream WAL to MinIO.

13.3 Recovery Priorities

Priority	Data	RTO
Critical	Vaultwarden (password vault)	Restore from Velero backup
High	Infisical secrets, CNPG data	Restore from WAL / Velero
High	Immich photo library (200 Gi PVC)	Restore PVC from Ceph snapshot
Medium	Nextcloud files, MongoDB	Restore from Velero
Low	Redis, monitoring TSDB	Ephemeral acceptable, re-seed

Appendix: Known Operational Notes

Infisical Webhook (CRITICAL)

The infisical bundled nginx creates infisical-ingress-nginx-admission ValidatingWebhookConfiguration which blocks ALL ingress and ExternalSecret creation cluster-wide when it times out.

Permanent fix applied: infisical nginx scaled to 0, admissionWebhooks.enabled: false, failurePolicy: Ignore patched.

If it reappears:

```
kubectl delete validatingwebhookconfiguration infisical-ingress-nginx-admission 2>/dev/null; true
kubectl delete service infisical-ingress-nginx-controller-admission -n infisical 2>/dev/null; true
```

ESO Webhook Disabled

ESO webhook and certController are both disabled. Do not re-enable without adding a CiliumNetworkPolicy allowing kube-apiserver node IP → ESO webhook service.

Ceph OSD Egress

Standard NetworkPolicy cannot reach Ceph OSDs on ports 6802-6809 (bare-metal IPs) in Cilium native routing mode. Use CiliumNetworkPolicy with toCIDR: 192.168.22.0/24 for any namespace mounting Ceph RBD volumes.

Infisical Ingress Class Reversion

After ArgoCD sync, infisical ingress may revert to infisical-nginx class. Manual patch:

```
kubectrl patch ingress infisical-ingress -n infisical --type='json' \
-
p='[{"op":"replace","path":"/spec/ingressClassName","value":"nginx"},...]'
```